

CBSE Class 12 Chemistry — Aldehydes, Ketones and Carboxylic Acids

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

Q1. Which of the following compounds will NOT undergo the aldol condensation reaction? (a) CH_3CHO (b) CH_3CH_2CHO (c) CH_3COCH_3 (d) $HCHO$ [PYQ 2021] [Confidence: Very High]

Q2. The Iodoform test is NOT given by which of the following compounds? (a) 2-Pentanone (b) Ethanol (c) Ethanal (d) 3-Pentanone [PYQ 2020] [Confidence: Very High]

Q3. Which of the following tests/reactions is given by aldehydes as well as ketones? (a) Fehling's test (b) Tollen's test (c) 2,4-DNP test (d) Cannizzaro reaction [PYQ SQP 2022] [Confidence: Very High]

Q4. The appropriate reagent for the transformation of an amide to an amine via degradation is: (a) NH_2NH_2, OH^- (b) $Br_2, NaOH$ (c) H_2/Ni (d) $LiAlH_4$ [Practice] [Confidence: High]

Q5. Which of the following compounds will undergo the Cannizzaro reaction? (a) CH_3CHO (b) CH_3COCH_3 (c) C_6H_5CHO (d) $C_6H_5CH_2CHO$ [PYQ 2021] [Confidence: Very High]

Q6. The slow and rate-determining step in the mechanism of the Cannizzaro reaction is: (a) Attack of nucleophile (b) Hydride transfer (c) Proton transfer (d) Deprotonation [PYQ 2021] [Confidence: High]

Q7. Which of the following is the correct order of acidic strength? (a) $CF_3COOH > CHCl_2COOH > HCOOH > C_6H_5CH_2COOH > CH_3COOH$ (b) $CH_3COOH > HCOOH > CF_3COOH > CHCl_2COOH > C_6H_5CH_2COOH$ (c) $HCOOH > C_6H_5CH_2COOH > CF_3COOH > CHCl_2COOH > CH_3COOH$ (d) $CF_3COOH > CH_3COOH > HCOOH > CHCl_2COOH > C_6H_5CH_2COOH$ [PYQ 2021] [Confidence: Very High]

Q8. Assertion (A): Carboxylic acids are more acidic than phenols. Reason (R): Phenols are ortho and para directing. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2020] [Confidence: Very High]

Q9. Assertion (A): $(CH_3)_3CCOOH$ does not give the Hell-Volhard-Zelinsky (H.V.Z) reaction. Reason (R): $(CH_3)_3CCOOH$ does not have an α -hydrogen atom. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2021] [Confidence: Very High]

Q10. Assertion (A): Strong oxidising agents oxidise toluene and its derivatives to benzoic acids. Reason (R): It is possible to stop the oxidation of toluene at the aldehyde stage with suitable reagents. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ SQP 2023] [Confidence: High]

Q11. Assertion (A): α -Hydrogen atoms in aldehydes and ketones are exceptionally acidic. Reason (R): The anion left after the removal of α -hydrogen is highly stabilised by resonance. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2021] [Confidence: Very High]

Q12. What is the IUPAC name of the ketone which undergoes the iodoform reaction to give $CH_3CH = C(CH_3)COONa$ and a yellow precipitate of CHI_3 ? (a) 3-Methylpent-3-en-2-one (b) 3-Methylbut-2-en-one (c) 4-Methylpent-3-en-2-one (d) 3-Methylpent-4-one [PYQ SQP 2023] [Confidence: High]

Q13. Write the structure of 3-methylpentanal. [PYQ 2015] [Confidence: Very High]

Q14. Identify the reagent used to selectively reduce a nitrile group ($-C \equiv N$) to an aldehyde using $SnCl_2$ and HCl followed by hydrolysis. [PYQ 2015] [Confidence: Very High]

Q15. Which derivative is formed when an aldehyde or ketone reacts with 2,4-dinitrophenylhydrazine (2,4-DNP)? [PYQ 2025-26 SQP] [Confidence: High]

Q16. Write the IUPAC name of CH_3CH_2CHO . [PYQ 2015] [Confidence: Very High]

Q17. Out of propanal and propanone, which one will give a positive Fehling's test? [Practice] [Confidence: Very High]

Q18. Name the specific reaction where benzaldehyde is treated with concentrated NaOH to yield benzyl alcohol and sodium benzoate. [PYQ 2021] [Confidence: Very High]

Q19. Which acid is stronger: Acetic acid or Chloroacetic acid? [PYQ 2021] [Confidence: Very High]

Q20. What is the formal oxidation state of the carbonyl carbon in formaldehyde ($HCHO$)? [Practice] [Confidence: Medium]

2-Mark Questions (Short Answer)

Q21. Give simple chemical tests to distinguish between the following pairs of compounds: (A) Benzaldehyde and Benzoic acid (B) Propanal and Propanone [PYQ 2014] [Confidence: Very High]

Q22. Arrange the following compounds in the increasing order of their boiling points: CH_3CHO , CH_3COOH , CH_3CH_2OH [PYQ 2016] [Confidence: Very High]

Q23. Arrange the following in the increasing order of their reactivity towards nucleophilic addition reactions: $HCHO$, CH_3CHO , CH_3COCH_3 , $C_6H_5COCH_3$ [PYQ 2016]

[Confidence: Very High]

Q24. Explain why aromatic carboxylic acids do not undergo Friedel-Crafts alkylation or acylation reactions. [PYQ 2018] [Confidence: Very High]

Q25. Carboxylic acids contain a carbonyl group but do not give the characteristic nucleophilic addition reactions of aldehydes and ketones. Give a reason. [PYQ 2017] [Confidence: Very High]

Q26. Write the chemical equations involved in the following reactions: (A) Clemmensen reduction of acetone. (B) Etard reaction of toluene. [PYQ Term-2 2022 / 2015] [Confidence: Very High]

Q27. How do you convert the following? (A) Ethyne to ethanal (B) Toluene to benzoic acid [PYQ 2015 / 2018] [Confidence: High]

Q28. Write the main product(s) formed in the following reactions: (A) Propanone is treated with methyl magnesium bromide (CH_3MgBr) and then hydrolysed (H_3O^+). (B) Acetophenone is treated with $Zn(Hg)$ and concentrated HCl . [PYQ 2020 / 2019] [Confidence: Very High]

Q29. Give simple chemical tests to distinguish between the following pairs of compounds: (A) Phenol and Benzoic acid (B) Pentan-2-one and Pentan-3-one [PYQ Term-2 2022 / Practice] [Confidence: Very High]

Q30. Write the chemical equation to illustrate the Hell-Volhard-Zelinsky (HVZ) reaction using propanoic acid. [PYQ 2014] [Confidence: Very High]

Q31. Account for the following: The pK_a value of 4-nitrobenzoic acid is lower than that of benzoic acid. [PYQ 2018] [Confidence: Very High]

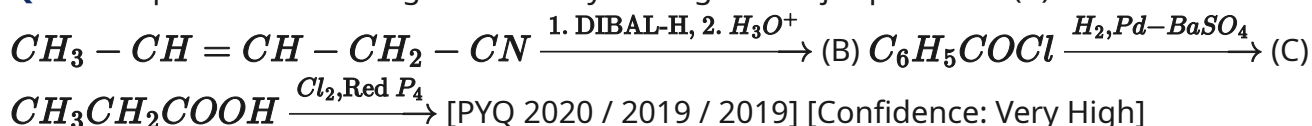
Q32. Draw the structures of the following derivatives: (A) Propanone oxime (B) Semicarbazone of ethanal [PYQ 2015] [Confidence: High]

3-Mark Questions (Short Answer)

Q33. How can you bring about the following conversions? (A) Benzene to acetophenone (B) Propanone to propene (C) Benzoic acid to benzaldehyde [PYQ 2020 / 2020 / Delhi Gov SQP 2022] [Confidence: Very High]

Q34. Account for the following observations: (A) The α -hydrogen atoms in aldehydes are acidic in nature. (B) Oxidation of propanone requires drastic conditions and yields a mixture of carboxylic acids. (C) Benzaldehyde is less reactive than propanal toward nucleophilic addition reactions. [Practice / PYQ themes] [Confidence: Very High]

Q35. Complete the following reactions by writing the major products: (A)



Q36. Arrange the following sets of compounds as directed: (A) Ethanal, propanone, propanal, butanone (Increasing reactivity towards nucleophilic addition). (B) 4-Nitrobenzoic acid, benzoic acid, 3,4-dinitrobenzoic acid, 4-methoxybenzoic acid (Increasing acid strength). (C) Acetaldehyde, Acetone, Di-tert-butyl ketone (Increasing steric hindrance to nucleophilic attack). [CBSE Term-2 2022] [Confidence: High]

Q37. Describe the following reactions with a suitable chemical equation for each: (A) Aldol condensation (B) Decarboxylation (C) Cross Aldol Condensation [PYQ 2015 / Practice] [Confidence: Very High]

Q38. Compound A and B are two functional isomers of molecular formula C_3H_6O . On heating with NaOH and I_2 , isomer B forms a yellow precipitate of iodoform, whereas isomer A does not form any precipitate. (A) Identify the functional groups and write the formulae of A and B. (B) Write the chemical equation for the reaction of B with I_2 and NaOH. [PYQ 2016] [Confidence: Very High]

Q39. How will you convert the following in not more than two steps: (A) Ethylbenzene to benzoic acid (B) Acetophenone to benzoic acid (C) Ethanoic acid to 2-hydroxyethanoic acid [PYQ 2017] [Confidence: High]

Q40. X, Y, and Z are three non-cyclic functional isomers of a carbonyl compound with the molecular formula C_4H_8O . Isomers X and Z give a positive Tollen's test, whereas isomer Y does not give Tollen's test but gives a positive Iodoform test. Isomers X and Y on reduction with $Zn(Hg)/conc. HCl$ give the same product (W). Write the structures of X, Y, Z, and W. [PYQ 2018] [Confidence: Very High]

Q41. Predict the products of the following reactions: (A) $C_6H_5 - CO - CH_3 \xrightarrow{NaOH/I_2}$ (B) $C_6H_5 - CHO + H_2N - NH - C_6H_5 \xrightarrow{H^+}$ (C) $CH_3 - COOH \xrightarrow{PCl_5}$ [PYQ 2015 / Practice] [Confidence: High]

Q42. Write the product formed when benzaldehyde reacts with the following reagents: (A) CH_3CHO in the presence of dilute NaOH (B) Tollen's reagent (C) Concentrated NaOH [CBSE SQP Term-2 2022] [Confidence: Very High]

5-Mark Questions (Long Answer / Case-Study)

Q43. An organic compound A (C_8H_6) on treatment with dilute sulphuric acid containing mercuric sulphate gives a compound B. Compound B can also be obtained from a reaction of benzene with an acid chloride in the presence of anhydrous aluminium chloride. Compound B, when treated with iodine in aqueous KOH, yields C and a yellow compound D. (A) Identify A, B, C, and D with justification. (4 Marks) (B) Show the chemical equation for how B is formed from A. (1 Mark) [PYQ 2020] [Confidence: Very High]

Q44. Case Study: Aldol Condensation The addition reaction of an enol or enolate to the carbonyl functional group of an aldehyde or ketone is known as aldol addition. The β -hydroxyaldehyde or β -hydroxyketone so obtained undergoes dehydration in the second step

to produce a conjugated enone. Carbonyl compounds having an α -hydrogen undergo aldol condensation reactions. (A) Which combination of carbonyl compounds gives phenyl vinyl ketone by cross-aldol condensation? (1 Mark) (B) Write the IUPAC name and structure of the main product formed by the aldol condensation of acetaldehyde and acetone. (2 Marks) (C) What is the reverse reaction of the condensation step called? (1 Mark) (D) Draw the structure of the aldol formed specifically from the self-condensation of propanal. (1 Mark) [PYQ 2021 / MTG] [Confidence: High]

Q45. Case Study: Cannizzaro Reaction When an aldehyde with no α -hydrogen reacts with concentrated aqueous NaOH, half the aldehyde is converted to a carboxylic acid salt and the other half is converted to an alcohol. This disproportionation reaction is known as the Cannizzaro reaction. (A) A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives specific products. Name them and explain why formaldehyde is preferentially oxidized. (2 Marks) (B) Trichloroacetaldehyde is subjected to the Cannizzaro reaction using NaOH. The mixture of the products contains the sodium trichloroacetate ion and another compound. Name the other compound. (1 Mark) (C) Identify the slowest (rate-determining) step in the mechanism of the Cannizzaro reaction. (1 Mark) (D) Write the complete balanced Cannizzaro reaction for Benzaldehyde. (1 Mark) [PYQ 2021 / MTG] [Confidence: Very High]

Q46. A hydrocarbon (A) with molecular formula C_5H_{10} on ozonolysis gives two products (B) and (C). Both (B) and (C) give a yellow precipitate when heated with iodine in the presence of NaOH, while only (B) gives a silver mirror on reaction with Tollen's reagent. (A) Identify the structural formulas of (A), (B), and (C). (3 Marks) (B) Write the reaction of (B) with Tollen's reagent. (1 Mark) (C) Write the equation for the iodoform test for (C). (1 Mark) [CBSE SQP 2022] [Confidence: Very High]

Q47. An organic compound (A) with molecular formula $C_2Cl_3O_2H$ is obtained when compound (B) reacts with Red Phosphorus and Cl_2 . The organic compound (B) can be obtained from the reaction of methyl magnesium chloride with dry ice (CO_2) followed by acid hydrolysis. (A) Identify compounds (A) and (B). (2 Marks) (B) Write down the reaction for the formation of (A) from (B). What is the specific name of this reaction? (2 Marks) (C) Which will be the more acidic compound, (A) or (B)? Briefly justify why. (1 Mark) [CBSE SQP 2022] [Confidence: Very High]

Q48. An organic compound A (C_3H_6O) is highly resistant to oxidation but forms a compound B (C_3H_8O) on catalytic reduction. Compound B reacts with HBr to form a bromide C, which reacts with KCN to form D. Compound D on acidic hydrolysis gives a carboxylic acid E. (A) Identify the structures and IUPAC names of compounds A, B, C, D, and E. (5 Marks) [Practice] [Confidence: High]

Q49. Account for the following observations in organic chemistry: (A) Carboxylic acids do not give the characteristic nucleophilic addition reactions of the carbonyl group (e.g., forming oximes or hydrazones). (1 Mark) (B) Electrophilic substitution in benzoic acid takes place at the meta position. (1 Mark) (C) The α -hydrogen atoms of aldehydes and ketones are acidic in nature. (1 Mark) (D) Propanone is less reactive than ethanal toward nucleophilic addition

reactions. (1 Mark) (E) Carboxylic acids have significantly higher boiling points than alcohols of comparable molecular mass. (1 Mark) [PYQ Themes 2014-2018] [Confidence: Very High]

Q50. Two moles of an organic compound 'A' on treatment with a strong base (like 50% NaOH) undergo disproportionation to give two compounds 'B' and 'C'. Compound 'B' on dehydrogenation with heated Cu gives 'A' back. Acidification of 'C' yields a carboxylic acid 'D' with the molecular formula CH_2O_2 . (A) Identify the exact structures and names of A, B, C, and D. (4 Marks) (B) Name the specific chemical reaction involved when A converts to B and C. (1 Mark) [Reasoning PYQ Themes] [Confidence: High]

Answer Key

Q1. (d) $HCHO$ (Formaldehyde lacks an α -hydrogen). **Q2.** (d) 3-Pentanone (Lacks a terminal methyl ketone group CH_3CO-). **Q3.** (c) 2,4-DNP test (Both form 2,4-dinitrophenylhydrazones). **Q4.** (b) $Br_2, NaOH$ (Hofmann bromamide degradation). **Q5.** (c) C_6H_5CHO (Benzaldehyde lacks an α -hydrogen). **Q6.** (b) Hydride transfer (from the tetrahedral intermediate to the second aldehyde molecule). **Q7.** (a) $CF_3COOH > CHCl_2COOH > HCOOH > C_6H_5CH_2COOH > CH_3COOH$ (-I effect strength). **Q8.** (b) Both A and R are true but R is not the correct explanation of A. (Reason for higher acidity is the resonance stabilization of the carboxylate ion over phenoxide). **Q9.** (a) Both A and R are true and R is the correct explanation of A. **Q10.** (b) Both A and R are true but R is not the correct explanation of A. (The reason is a separate fact, not the explanation for why strong oxidizers yield benzoic acid). **Q11.** (a) Both A and R are true and R is the correct explanation of A. (Resonance stabilization of the enolate ion). **Q12.** (c) 4-Methylpent-3-en-2-one (Often called mesityl oxide; $CH_3C(CH_3)=CH-CO-CH_3$). **Q13.** $CH_3-CH_2-CH(CH_3)-CH_2-CHO$ **Q14.** Stephen Reduction ($SnCl_2 / HCl$). **Q15.** 2,4-dinitrophenylhydrazone (a yellow, orange, or red solid). **Q16.** Propanal. **Q17.** Propanal (Aliphatic aldehydes give a positive Fehling's test, ketones do not). **Q18.** Cannizzaro reaction. **Q19.** Chloroacetic acid (due to the -I effect of the chlorine atom stabilizing the carboxylate anion). **Q20.** 0 (Carbon is bonded to two Hydrogens (+1 each) and one Oxygen (-2)).

Q21. (A) Add $NaHCO_3$: Benzoic acid gives brisk effervescence of CO_2 , benzaldehyde does not. (B) Tollen's test: Propanal gives a silver mirror, propanone does not. **Q22.**

$CH_3CHO < CH_3CH_2OH < CH_3COOH$ (Dipole interactions < H-bonding in alcohol < Stronger extensive H-bonding forming dimers in acid). **Q23.**

$C_6H_5COCH_3 < CH_3COCH_3 < CH_3CHO < HCHO$ (Steric hindrance and +I / +R effects decrease reactivity). **Q24.** The carboxyl group ($-COOH$) is strongly deactivating and electron-withdrawing. It acts as a Lewis base and bonds with the Lewis acid catalyst ($AlCl_3$), halting the reaction. **Q25.** The carbonyl carbon in carboxylic acids is less electrophilic due to resonance delocalization of the lone pair of electrons from the hydroxyl oxygen atom. **Q26.**

(A) $CH_3COCH_3 + 4[H] \xrightarrow{Zn-Hg/conc. HCl} CH_3CH_2CH_3 + H_2O$. (B)

$C_6H_5CH_3 + 2CrO_2Cl_2 \xrightarrow{CS_2} \text{Complex} \xrightarrow{H_3O^+} C_6H_5CHO$. **Q27.** (A) Pass ethyne through dil. H_2SO_4 and $HgSO_4$ at 333K (hydration). (B) Heat toluene with alkaline $KMnO_4$, followed by acidification (H_3O^+). **Q28.** (A) 2-Methylpropan-2-ol (tert-butyl alcohol). (B) Ethylbenzene ($C_6H_5CH_2CH_3$). **Q29.** (A) $FeCl_3$ test: Phenol gives a violet color, benzoic acid gives a buff-colored precipitate. (B) Iodoform test: Pentan-2-one gives a yellow precipitate of CHI_3 , Pentan-3-one does not. **Q30.**

$CH_3CH_2COOH + Cl_2 \xrightarrow{\text{Red P, } H_2O} CH_3CH(Cl)COOH + HCl$. **Q31.** The $-NO_2$ group exerts a strong electron-withdrawing effect (-I, -R), which stabilizes the conjugate base (4-nitrobenzoate ion) more effectively than in benzoic acid. **Q32.** (A) $(CH_3)_2C = N - OH$. (B) $CH_3CH = N - NH - CONH_2$.

Q33. (A) Treat benzene with acetyl chloride (CH_3COCl) in the presence of anhydrous $AlCl_3$ (Friedel-Crafts acylation). (B) Reduce propanone to propan-2-ol ($NaBH_4$), then dehydrate with conc. H_2SO_4 at 443K. (C) Convert benzoic acid to benzoyl chloride ($SOCl_2$), then reduce via Rosenmund reduction ($H_2/Pd - BaSO_4$). **Q34.** (A) The resulting enolate anion is highly stabilized by resonance. (B) Ketones lack an oxidizable C-H bond on the carbonyl carbon; oxidation requires breaking strong C-C bonds. (C) The +R (resonance) effect of the benzene ring reduces the electrophilicity of the carbonyl carbon. **Q35.** (A)

$CH_3 - CH = CH - CH_2 - CHO$. (B) C_6H_5CHO (Benzaldehyde). (C) $CH_3CH(Cl)COOH$ (2-Chloropropanoic acid). **Q36.** (A) Butanone < Propanone < Propanal < Ethanal. (B) 4-Methoxybenzoic acid < Benzoic acid < 4-Nitrobenzoic acid < 3,4-Dinitrobenzoic acid. (C) Acetaldehyde < Acetone < Di-tert-butyl ketone. **Q37.** (A)

$2CH_3CHO \xrightarrow{\text{dil. } NaOH} CH_3CH(OH)CH_2CHO \xrightarrow{\Delta} CH_3CH = CHCHO$. (B)

$CH_3COONa + NaOH \xrightarrow{CaO, \Delta} CH_4 + Na_2CO_3$. (C) Reaction between two different aldehydes/ketones with α -H (e.g., ethanal + propanal + dil. NaOH $\rightarrow$ mixture of 4 products).

Q38. (A) A is Propanal (CH_3CH_2CHO), B is Propanone (CH_3COCH_3). (B)

$CH_3COCH_3 + 3I_2 + 4NaOH \rightarrow CHI_3 \downarrow (\text{yellow}) + CH_3COONa + 3NaI + 3H_2O$.

Q39. (A) Oxidize ethylbenzene with alkaline $KMnO_4$ and heat, then acidify. (B) Oxidize acetophenone with alkaline $KMnO_4$ /heat, then acidify (or use Iodoform reaction followed by acidification). (C) React ethanoic acid with Cl_2 /Red P (HVZ) to get chloroethanoic acid, then boil with aq. KOH. **Q40.** X is Butanal ($CH_3CH_2CH_2CHO$); Z is 2-Methylpropanal ($(CH_3)_2CHCHO$); Y is Butan-2-one ($CH_3COCH_2CH_3$); W is Butane ($CH_3CH_2CH_2CH_3$). **Q41.** (A) $C_6H_5COONa + CHI_3 \downarrow$. (B) $C_6H_5CH = N - NH - C_6H_5$ (Benzaldehyde phenylhydrazone). (C) $CH_3COCl + POCl_3 + HCl$. **Q42.** (A) $C_6H_5CH = CHCHO$ (Cinnamaldehyde, via cross-aldol). (B) $C_6H_5COO^- + Ag \downarrow$ (Silver mirror). (C)

$C_6H_5CH_2OH + C_6H_5COONa$ (Cannizzaro products).

Q43. (A) A is Ethynylbenzene (Phenylacetylene, $C_6H_5C \equiv CH$). B is Acetophenone ($C_6H_5COCH_3$). C is Potassium benzoate (C_6H_5COOK). D is Iodoform (CHI_3). (B)

$C_6H_5C \equiv CH + H_2O \xrightarrow{Hg^{2+}, H^+} C_6H_5C(OH) = CH_2 \rightleftharpoons C_6H_5COCH_3$. **Q44.** (A)

Acetophenone and Formaldehyde (cross-aldol). (B) 4-Hydroxypentan-2-one ($CH_3CH(OH)CH_2COCH_3$) or 3-Hydroxybutanal depending on which acts as nucleophile.

Main dehydration product is 4-methylpent-3-en-2-one. (C) Retro-aldol (hydrolysis/cleavage). (D) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{CH}_3)\text{CHO}$. **Q45.** (A) Benzyl alcohol ($\text{C}_6\text{H}_5\text{CH}_2\text{OH}$) and Sodium formate (HCOONa). Formaldehyde is more reactive and easily oxidized to formate, reducing benzaldehyde to alcohol (Crossed Cannizzaro). (B) 2,2,2-Trichloroethanol. (C) The transfer of a hydride ion (H^-) from the tetrahedral intermediate to the second aldehyde molecule. (D) $2\text{C}_6\text{H}_5\text{CHO} + \text{conc. NaOH} \rightarrow \text{C}_6\text{H}_5\text{CH}_2\text{OH} + \text{C}_6\text{H}_5\text{COONa}$. **Q46.** (A) A is 2-Methylbut-2-ene ($\text{CH}_3\text{C}(\text{CH}_3) = \text{CHCH}_3$). B is Ethanal (CH_3CHO). C is Propanone (CH_3COCH_3). (B) $\text{CH}_3\text{CHO} + 2[\text{Ag}(\text{NH}_3)_2]^+ + 3\text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + 2\text{Ag} \downarrow + 4\text{NH}_3 + 2\text{H}_2\text{O}$. (C) $\text{CH}_3\text{COCH}_3 + 3\text{I}_2 + 4\text{NaOH} \rightarrow \text{CHI}_3 \downarrow + \text{CH}_3\text{COONa} + 3\text{NaI} + 3\text{H}_2\text{O}$. **Q47.** (A) A is Trichloroacetic acid (CCl_3COOH). B is Acetic acid (CH_3COOH). (B) $\text{CH}_3\text{COOH} + 3\text{Cl}_2 \xrightarrow{\text{Red P}} \text{CCl}_3\text{COOH} + 3\text{HCl}$. This is the Hell-Volhard-Zelinsky (HVZ) reaction. (C) Compound A (CCl_3COOH) is significantly more acidic due to the strong -I (electron-withdrawing) effect of three chlorine atoms stabilizing the carboxylate ion. **Q48.** A is Propanone (CH_3COCH_3). B is Propan-2-ol ($\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$). C is 2-Bromopropane ($\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$). D is Isobutyronitrile / 2-Methylpropanenitrile ($(\text{CH}_3)_2\text{CHCN}$). E is Isobutyric acid / 2-Methylpropanoic acid ($(\text{CH}_3)_2\text{CHCOOH}$). **Q49.** (A) The carbonyl carbon in carboxylic acids is stabilized by resonance from the -OH oxygen, severely lowering its electrophilicity. (B) The carboxyl group is electron-withdrawing (-R), which deactivates the ring and specifically depletes electron density at the ortho and para positions, leaving the meta position relatively electron-rich for attack. (C) The resulting enolate carbanion is highly stabilized by resonance delocalization onto the electronegative carbonyl oxygen. (D) Propanone has two electron-donating methyl groups (+I effect) that reduce the positive charge on the carbonyl carbon and cause greater steric hindrance compared to the single methyl group in ethanal. (E) Carboxylic acids form extensive and very strong intermolecular hydrogen bonds, existing largely as stable dimers even in the vapor phase. **Q50.** (A) A is Formaldehyde (HCHO). B is Methanol (CH_3OH). C is Sodium formate (HCOONa). D is Formic acid / Methanoic acid (HCOOH). (B) Cannizzaro reaction.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).